Obsessive-Compulsive Disorder

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CPhA strives to provide guidance for health-care professionals on the appropriate care of all individuals. However, we recognize that in this chapter, our content is based on clinical research studies of patients who may not reflect the identity of the individual patient presenting. Other resources exist that may be helpful, such as the World Professional Association for Transgender Health's (WPATH) Standard of Care. When the terms "male" and "female" are used in the text, it is used to maintain accuracy with what is reported in the clinical literature. We are aware these terms are not inclusive of all patients.

Introduction

Obsessive-compulsive disorder (OCD) is classified separately from anxiety disorders in the Diagnostic and Statistical Manual of Mental Disorders 5th edition, text revision (DSM-5-TR).[1] Related conditions in the same classification include body dysmorphic disorder, hoarding disorder, trichotillomania (hair-pulling disorder), excoriation disorder, and OCD that is substance/medication-induced or related to another medical condition.

OCD frequently starts early in life and often becomes a chronic condition if inadequately treated. The mean age of onset is about 20 years with 25% of cases beginning by age 14. Although the onset is typically earlier in males, OCD is 1.6 times more common in females.[2] The defining symptoms are obsessions and compulsions. Obsessions are thoughts, images or urges that provoke anxiety and are unwanted, repetitive and/or difficult to control. Compulsions consist of repetitive behaviours that may be visible actions, such as washing or checking, or may be mental actions such as counting or repeating a phrase in a precise manner. In severe OCD, these ritualistic behaviours occupy many hours each day and can be extremely disabling.

Goals of Therapy

- Eliminate or decrease symptoms of OCD
- Eliminate or decrease OCD-associated disability
- Prevent recurrence
- Restore quality of life
- Treat comorbid conditions

Investigations

- Thorough history with attention to:
  - nature and onset of symptoms
  - nature and extent of disability
  - related difficulties (e.g., beliefs), psychosocial functioning and quality of life
  - presence of comorbid medical or psychiatric conditions
    - comorbid mood disorder should be differentiated from secondary depression that commonly occurs in OCD and spontaneously remits with improvement in OCD symptoms
- Criteria and interview questions assist in obtaining an accurate diagnosis (see Table 1, Table 2) as well as standardized measures of symptoms, e.g., Yale-Brown Obsessive Compulsive Scale (Y-BOCS)[3][4]

Rarely, obsessive-compulsive symptoms may be triggered by medical conditions or drugs. History should involve screening for head injuries, neurocognitive disorders, new use of stimulants, dopamine agonists (e.g., pramipexole) and serotonin-dopamine antagonists (e.g., clozapine).[5] Pediatric autoimmune neuropsychiatric disorders associated with streptococcus (PANDAS) is a controversial topic in the medical literature suggesting that a recent streptococcal infection may trigger the onset of OCD symptoms in children and adults.[6]

***Table 1:** Diagnostic Criteria*[7]

• Presence of obsessions (persistent, disturbing thoughts that cannot be reasoned away) or compulsions (uncontrollable impulses to do something against one's conscious will)
• Patient has recognized that the obsessions or compulsions are excessive or unreasonable
• Obsessions/compulsions cause distress, are time-consuming, or interfere with patient's routine, occupation, or academic or social functioning

	• Obsessions/compulsions are not due to substance abuse or another medical or mental disorder

Table 2: Interview Questions to Identify Presence of Obsessions or Compulsions[7]

Obsessions	Is patient experiencing disturbing thoughts, images or urges that recur or are difficult to ignore? For example, "Do troubling thoughts or images come into your mind without you wanting to have them and are they difficult to get rid of?"
Compulsions	Does patient feel compelled to do something that doesn't make sense to them or that the person doesn't want to do? For example, "Do you feel that you have to count, wash, clean or check things repeatedly when you know that it isn't really necessary?"

Therapeutic Choices

Specialized **cognitive behavioural therapy** (CBT) and pharmacotherapy with selective serotonin reuptake inhibitors (**SSRIs**) are first-line treatments for OCD. The efficacy for both modalities is supported by a significant number of high-quality studies. Consider CBT as first-line therapy for most patients, since pharmacotherapy is associated with a high risk of relapse when treatment is discontinued. However, since access to therapists with expertise in OCD is limited, starting with antidepressants until CBT can be obtained is a reasonable treatment approach. An important advance in this field is the development of knowledge and competency standards for specialized treatments for OCD throughout the lifespan.[8]

Many OCD pharmacotherapy trials define clinically meaningful response as at least a 30% reduction in the Y-BOCS. There is increasing focus in the CBT literature on rates of wellness and recovery following specialized treatment. Recovery from illness is usually defined as scores falling in the range of the general population.[9]

Nonpharmacologic Choices

The first-line psychotherapeutic treatment of choice for OCD is specialized CBT, including **exposure and response prevention** (ERP).[10][11][12] Specialized CBT refers to interventional specificity for varied OCD subtypes that differs substantively from CBT developed for other disorders, e.g., anxiety disorders. Meta-analysis of controlled trials of specialized CBT showed that approximately 65% of patients reported substantial reduction in symptoms (mean improvement of 50% at 15 months);[13] reported remission rates of 1669 participants were 59% post-treament (average 14–15 wk) and 57% at follow-up (average 15 months).[14] Patients with complex metacognitive dysfunction (e.g., symptom-related beliefs) and varied emotional responses may be difficult to treat with ERP alone and may require additional specialized cognitive therapy interventions.[15] Evidence-based specialized protocols have been developed for different OCD subtypes that respond differentially to specific CBT interventions.[16]

Additionally, many OCD patients do not respond to weekly in-office sessions and require increased session frequency/duration and therapist-assisted specialty protocols in their naturalistic environments. OCD centres that offer subtype-specific interventions, longer-term therapy that is not time-limited, and therapist-assisted specialty CBT protocols report superior therapeutic outcomes and rates of full recovery.[13][17][18] Professionals in regions where this expertise is unavailable should refer to or consult with an expert in specialized CBT for OCD. Elaboration of specialized CBT protocols is beyond the scope of this chapter; see Suggested Readings for more information.

Low-intensity CBT (e.g., guided self-help) has not been found to reliably result in clinically significant improvement compared with wait-list control.[19] Internet-based CBT is promising for increased access to care and merits further study.[20] Importantly, stepped-care CBT for OCD (i.e., starting with self-help and progressing to specialized CBT) is not evidence-based and poses the serious risk of undertreatment, often resulting in avoidable symptom exacerbation and potential disability.[21][22]

OCD-related disorders (e.g., hoarding disorder, body dysmorphic disorder) also have specialized CBT protocols designed specifically for each disorder.[23][24][25]

Experimental Treatments

Brain stimulation therapies and neuroablative techniques have been investigated in the treatment of OCD. Brain stimulation therapies, including **repetitive transcranial magnetic stimulation** (rTMS), **deep transcranial magnetic stimulation** (dTMS) and **deep brain stimulation** (DBS), have been used to treat OCD that has not responded to the usual pharmacologic and psychotherapeutic options. rTMS consists of the application of magnetic stimulation to superficial areas of the cerebral cortex. Most studies evaluated the effectiveness of treatment after only 4 weeks; however, there is preliminary evidence for sustained benefit for up to 12 weeks.[26][27] dTMS is a newer technology that can stimulate deeper cerebral regions. In a randomized placebo-controlled trial, 45% of OCD patients treated with dTMS responded, compared to 18% who received the sham treatment, after a 1-month follow-up assessment.[28] The only common side effect was mild headache. There are no clearly established guidelines on the role of rTMS and dTMS in the treatment of OCD. Both procedures are well tolerated but difficult to access and require 5 treatments per week for several weeks. These interventions are usually reserved for patients who have not benefited from at least 1 optimal trial of specialized CBT and pharmacological therapy.

DBS is a neurosurgical technique that involves inserting a small neurostimulator into a targeted brain area. Although there is no consensus on the optimal area in which to implant the stimulator device, DBS has proven beneficial for OCD in small randomized trials. A meta-analysis of heterogeneous studies involving different DBS sites in refractory patients demonstrated a 50% response rate compared to 18% with the sham treatment (response was defined as >35% improvement on the Y-BOCS).[29] Full remission was uncommon at 8%. Adverse effects from this procedure were mostly mild but common. Due to the limited number and low quality of available studies, this technique should be considered only for the most refractory patients.

Neuroablation techniques are also used in the management of treatment-refractory OCD; they entail creating an irreversible lesion in a cerebral structure that contributes to OCD.[30] Neuroablation has been studied in treatment refractory–OCD, mostly in small open-label trials; approximately 54–62% of patients respond to therapy.[31] The risk of serious adverse effects is estimated to be about 5–21% depending on the procedure.[32] More RCTs are needed to evaluate the true efficacy and safety of these procedures. Currently, neuroablation should be used for only the most refractory patients and only after consultation with independent experts who are not involved in the patient's care.

Electroconvulsive therapy is not helpful in treating OCD and is appropriate *only* for patients with severe comorbid mood disorders.

Pharmacologic Choices

Pharmacologic options for OCD are listed in Figure 1, Table 3 and Table 4.

The SSRIs **citalopram**, **escitalopram**, **fluoxetine**, **fluvoxamine**, **paroxetine** and **sertraline** are recommended as first-line treatment for OCD.[7][33] Roughly 80% of medication benefits appear in the initial 6 weeks of treatment, although it may take up to 12 weeks for maximum improvement.[33] Lack of response after 4 weeks predicts a low likelihood of improvement at 12 weeks. It has been reported that subjects who had <20% improvement on the Y-BOCS at 4 weeks had only a 20% chance of response at 12 weeks compared with 55% for those with >20% improvement.[34] There is no strong evidence to suggest that SSRIs vary in efficacy, but patients may respond to or tolerate one drug better than the others. Higher doses of SSRIs are often more effective at the cost of poorer tolerability.[35] Clinicians should start at the lowest dose for 1 week, gradually increasing to the maximum tolerated dose for at least 6 weeks before considering alternative therapy.[36] Doses of SSRIs higher than usual approved doses (e.g., sertraline up to 400 mg) are sometimes used by specialists, with clear informed patient consent.[37] If doses higher than recommended are required, carefully screen patients for risk of developing seizures or torsades de pointes (e.g., congenital long QT syndrome, congestive heart failure, bradyarrhythmia, hypokalemia or hypomagnesemia). If treatment with SSRIs is unsuccessful, options include switching to a different SSRI, supratherapeutic dosing of an SSRI or progressing to augmentation strategies.

Venlafaxine, **clomipramine** and **mirtazapine** have also been investigated as monotherapy for OCD. Both clomipramine and venlafaxine are believed to exert their therapeutic effects in OCD via serotonin reuptake inhibition. They may be more effective than SSRIs for comorbid chronic pain[38] and/or comorbid major depressive disorder,[39] possibly due to the additional norepinephrine reuptake inhibition. Clomipramine has similar effectiveness to SSRIs in OCD but has poorer tolerability due to anticholinergic and antihistaminergic effects; it is also lethal in overdose.[40] One study suggested that switching from an SSRI to venlafaxine may be less effective than switching to another SSRI.[41] Mirtazapine often causes problematic weight gain, but has a lower risk of causing sexual dysfunction. The evidence for using mirtazapine in monotherapy is weak; it may be considered for individuals who cannot tolerate an SSRI.[42] Mirtazapine may also be considered as an augmentation agent in combination with SSRIs.[43]

Aripiprazole (a dopamine receptor partial agonist) and **risperidone** (a dopamine receptor antagonist) are frequently used as augmentation agents in patients who do not achieve adequate improvement with SSRI monotherapy; it may benefit about 30% of patients.[44][45][46] Olanzapine and quetiapine have mixed data to support their use.[46] Generally, titrate these medications to the middle of their dosing range (e.g., aripiprazole 10–15 mg, risperidone 0.5–3 mg), which is higher than doses usually used for augmentation in major depressive disorder.[46] Monitor patients at 4 weeks to determine efficacy of augmentation of SSRI therapy. Adverse effects (e.g., metabolic syndrome) will affect the tolerability of these drugs and must be weighed against their potential benefits.

Preliminary studies demonstrated some benefit with glutamate-modulating drugs such as **lamotrigine**, **memantine**, **topiramate** and **amantadine** in combination with SSRIs for refractory OCD.[7][47] Memantine, lamotrigine and amantadine are generally well tolerated and may be reasonable next-step options for patients who cannot tolerate or do not benefit from dopamine antagonist augmentation.[48][49] Topiramate has poorer tolerability, but may be considered if a patient has comorbid migraines, binge eating disorder or drug-induced weight gain.[50][51][52]

Ketamine is being investigated for OCD and has shown promise, but is not yet used in clinical practice.[53][54] There is mixed evidence suggesting a potential benefit for N-acetylcysteine, a nutraceutical, as an add-on to serotonergic agents;[55] it may be more beneficial for compulsive symptoms than obsessions.[56]

Benzodiazepines and lithium are not helpful in treating OCD and are usually used only for patients with severe comorbid mood or anxiety disorders.

OCD-related disorders (e.g., hoarding disorder and body dysmorphic disorder) are treated with serotonergic medications; refractory cases may benefit from augmentation with dopamine antagonists or haloperidol.[57][58][59]

Consider monitoring response to treatment using the Y-BOCS or a similar, briefer rating scale (e.g., the 6-item Brief Obsessive Compulsive Scale [BOCS]) after 4 weeks, 12 weeks and at key clinical decision points.[60] If successful, most guidelines recommend continuation of treatment for at least 1 year.[7] Relapse rates are high following medication monotherapy discontinuation. The decision to continue medication long term should be individualized; it is prudent to wait for a period of prolonged psychosocial stability before discontinuation. When stopping any antidepressant, taper gradually to minimize discontinuation effects and warn patients to report any early signs of relapse.[7] For more information about antidepressant discontinuation syndrome, see Depression.

Most patients do not achieve minimal OCD symptoms with medications alone.[61][62] Whenever possible, specialized CBT should be combined with pharmacologic treatment to optimize outcomes.[63][64] For more information about CBT, see Nonpharmacologic Choices. Adding ERP to medication substantially improves response rate and reduces susceptibility to relapse compared with medication alone.[64][65][66][67] After a course of CBT, individuals taking serotonin reuptake inhibitors (SRIs) may maintain stability after the SRI is tapered to discontinuation, with careful monitoring afterward.[68]

Combination therapy may also help to increase clinical response in treatment-resistant individuals, such as those with presence of poor insight, overvalued ideas, high distress and intolerance of distress that persist despite provision of CBT.[69][70]

Choices during Pregnancy and Breastfeeding

OCD and Pregnancy

During pregnancy, obsessive-compulsive symptoms, particularly obsessional ideas about causing harm to the baby, may worsen and can be extremely disturbing. Although patients are not at higher risk of harming their children compared with the general population, these ideas are often accompanied by marked guilt and depression.[71] Many people with psychiatric disorders do not bring their distress to the attention of health-care providers.

Whenever possible, it is important to screen for the presence of OCD or other psychiatric symptoms before conception occurs. Screening can be repeated during the pregnancy and particularly postpartum. If a patient is suffering from marked psychological distress related to pregnancy and breastfeeding, it is imperative to screen for the presence of mood symptoms and suicidality. When a patient experiences severe psychiatric symptoms during pregnancy or postpartum, referral to a psychiatrist may be necessary.

Preconceptional treatment can be offered for disorders that are producing significant distress or are interfering with functioning. OCD with high fear of contamination may prevent a patient from entering settings where there is a perceived high risk of coming into contact with infections.

Management during Pregnancy

There is good evidence to show that psychological treatments can have beneficial effects for patients who persist with a treatment program. Specialized CBT can be administered throughout pregnancy.[72] If symptoms are severe and producing significant impairment, pharmacotherapy with **SSRIs** is appropriate.[72][73] General principles for management of depression during pregnancy are applicable to the management of OCD disorders.[74] Paroxetine has higher associated risks than other SSRIs. See also Depression for information on management of depression during pregnancy and breastfeeding.

OCD and Breastfeeding

In the postpartum period, patients with OCD may experience severe anxiety that can impede sleep and erode confidence in caring for the child. A patient with severe OCD can be so tormented by thoughts of potentially harming their child that they may refuse to be involved with caring for the child at all. In such cases, a psychiatric consultation is urgently required.

As in pregnancy, nonpharmacologic options should be used whenever possible in the postpartum period, particularly in breastfeeding patients. If drug therapy is necessary, consider **escitalopram** and **sertraline**, since both have minimal concentrations in breast milk.[74][75]

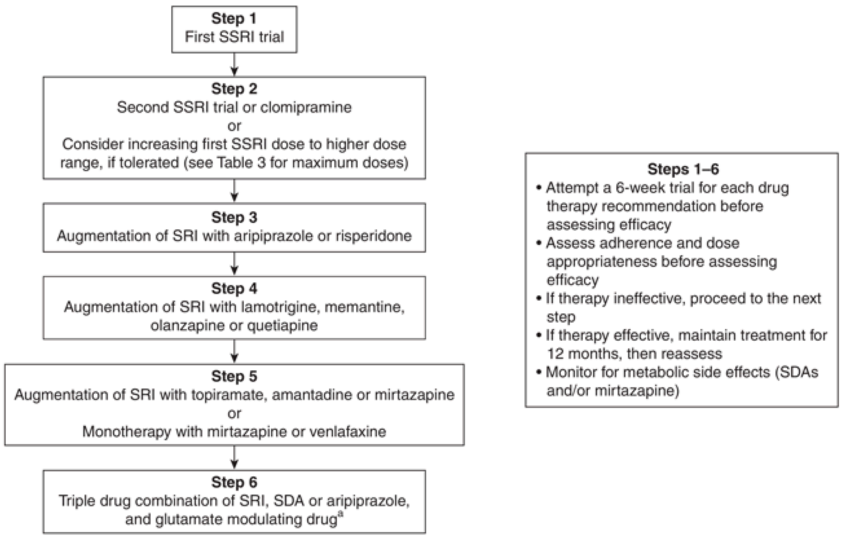
A discussion of general principles on the use of medications in these special populations can be found in Drugs Use during Pregnancy and Drug Use during Breastfeeding. Other specialized reference sources are also provided in these appendices.

Therapeutic Tips

- Administer specialized CBT specific for OCD as soon as possible.
- If the patient feels “improved” or “much improved” after 6 weeks of SSRI treatment, maintain the same treatment for a full 12-week trial. If the patient does not feel “improved” or “much improved” after 6 weeks, assess adherence and consider dose escalation to the maximum recommended/tolerated before switching agents. SSRI doses above the usual approved doses may be considered if the treatment is well tolerated at high doses (caution with citalopram and escitalopram due to risk of QTc prolongation).
- If the first SSRI at optimal dosage is not effective or tolerated, consider switching to another SSRI or clomipramine. If the first SSRI results in partial improvement, consider augmenting with risperidone or aripiprazole rather than switching to a second SRI. If the second SRI is not effective, consider augmenting with risperidone or aripiprazole rather than attempting a third SRI monotherapy trial. After this point, glutamate-modulating agents (lamotrigine or memantine) may also be tried.
- It is important to differentiate between technical treatment failure (no improvement due to inadequacy of treatment) and actual treatment failure (no improvement despite adequately delivered treatment).[76][77]

Algorithms

Figure 1: *Drug Therapy for Obsessive-Compulsive Disorder*



[a] Glutamate-modulating drugs refers to lamotrigine, memantine, topiramate and amantadine.

Abbreviations: SDA = serotonin/dopamine antagonist (referring to risperidone, olanzapine, quetiapine); SRI = serotonin reuptake inhibitor (referring to SSRIs, venlafaxine or clomipramine); SSRI = selective serotonin reuptake inhibitor

Drug Tables

Table 3: *Drugs for First-Line Management of Obsessive-Compulsive Disorder*[7]

Drug/Cost[a]	Dosage	Adverse Effects	Drug Interactions	Comments
Drug Class: Selective Serotonin Reuptake Inhibitors				
citalopram Celexa, CTP 30, generics < $30	Initial: 10–20 mg/day PO Usual: 20–40 mg/day PO Maximum: 40 mg/day PO; higher doses may be required[b][78]	Upon initiation: GI upset, anorexia, dry mouth, diaphoresis, headache, dizziness, insomnia, somnolence, anxiety, agitation, tremor. Usually resolve after 2 wk of therapy. Others: sexual dysfunction, weight gain, SIADH with hyponatremia. Citalopram: dose-related QTc prolongation.	SSRIs are all substrates of various CYP enzymes and subject to drug interactions. Concurrent use with MAOIs, linezolid and methylene blue is contraindicated due to increased risk of serotonin syndrome. Use caution if combining with other serotonergic agents. Increased risk of GI bleeding with NSAIDs, antiplatelet agents. Avoid concurrent use with drugs associated with prolonged QTc interval/torsades de pointes. Citalopram: higher risk of QTc prolongation, especially at high doses and in combination with other QTc-prolonging agents.	May take 2–3 months for maximum effect. Discontinue gradually.
escitalopram Cipralex, Kye-Escitalopram, other generics < $30	Initial: 5–10 mg/day PO Usual: 10–20 mg/day PO Maximum: 20 mg/day PO; higher doses may be required[b][78]	Upon initiation: GI upset, anorexia, dry mouth, diaphoresis, headache, dizziness, insomnia, somnolence, anxiety, agitation, tremor. Usually resolve after 2 wk of therapy. Others: sexual dysfunction, weight gain, SIADH with hyponatremia. Escitalopram: dose-related QTc prolongation.	SSRIs are all substrates of various CYP enzymes and subject to drug interactions. Concurrent use with MAOIs, linezolid and methylene blue is contraindicated due to increased risk of serotonin syndrome. Use caution if combining with other serotonergic agents. Increased risk of GI bleeding with NSAIDs, antiplatelet agents. Avoid concurrent use with drugs associated with prolonged QTc interval/torsades de pointes. Escitalopram: higher risk of QTc prolongation, especially at high doses and in combination with other QTc-prolonging agents.	May take 2–3 months for maximum effect. Discontinue gradually.

Drug/Cost[a]	Dosage	Adverse Effects	Drug Interactions	Comments
fluoxetine Prozac, Odan-Fluoxetine, other generics $30–60	Initial: 10–20 mg/day PO Usual: 40–80 mg/day PO Maximum: 80 mg/day PO	Upon initiation: GI upset, anorexia, dry mouth, diaphoresis, headache, dizziness, insomnia, somnolence, anxiety, agitation, tremor. Usually resolve after 2 wk of therapy. Others: sexual dysfunction, weight gain, SIADH with hyponatremia. Fluoxetine: anxiety upon initiation is common.	SSRIs are all substrates of various CYP enzymes and subject to drug interactions. Concurrent use with MAOIs, linezolid and methylene blue is contraindicated due to increased risk of serotonin syndrome. Use caution if combining with other serotonergic agents. Increased risk of GI bleeding with NSAIDs, antiplatelet agents. Avoid concurrent use with drugs associated with prolonged QTc interval/torsades de pointes. Fluoxetine: potent inhibitor of CYP2D6.	May take 2–3 months for maximum effect. Longer half-life (4–6 days) allows for a more rapid discontinuation than other SSRIs.
fluvoxamine Luvox, generics $30–60	Initial: 50–100 mg/day PO Usual: 200–300 mg/day PO Maximum: 300 mg/day PO	Upon initiation: GI upset, anorexia, dry mouth, diaphoresis, headache, dizziness, insomnia, somnolence, anxiety, agitation, tremor. Usually resolve after 2 wk of therapy. Others: sexual dysfunction, weight gain, SIADH with hyponatremia. Fluvoxamine: somnolence and nausea are common.	SSRIs are all substrates of various CYP enzymes and subject to drug interactions. Concurrent use with MAOIs, linezolid and methylene blue is contraindicated due to increased risk of serotonin syndrome. Use caution if combining with other serotonergic agents. Increased risk of GI bleeding with NSAIDs, antiplatelet agents. Avoid concurrent use with drugs associated with prolonged QTc interval/torsades de pointes. Fluvoxamine: potent inhibitor of CYP1A2.	May take 2–3 months for maximum effect. Discontinue gradually.
paroxetine immediate-release Paxil, generics < $30	Initial: 10–20 mg/day PO Usual: 40–60 mg/day PO Maximum: 60 mg/day PO	Upon initiation: GI upset, anorexia, dry mouth, diaphoresis, headache, dizziness, insomnia, somnolence, anxiety, agitation, tremor. Usually resolve after 2 wk of therapy. Others: sexual dysfunction, weight gain, SIADH with hyponatremia. Paroxetine: anticholinergic effects (dry mouth, constipation), somnolence are common.	SSRIs are all substrates of various CYP enzymes and subject to drug interactions. Concurrent use with MAOIs, linezolid and methylene blue is contraindicated due to increased risk of serotonin syndrome. Use caution if combining with other serotonergic agents. Increased risk of GI bleeding with NSAIDs, antiplatelet agents. Avoid concurrent use with drugs associated with prolonged QTc interval/torsades de pointes. Paroxetine: potent CYP2D6 inhibitor.	May take 2–3 months for maximum effect. Discontinue gradually.
paroxetine controlled-release Paxil CR $90–120	25–50 mg/day PO	Upon initiation: GI upset, anorexia, dry mouth, diaphoresis, headache, dizziness, insomnia, somnolence, anxiety, agitation, tremor. Usually resolve after 2 wk of therapy. Others: sexual dysfunction, weight gain, SIADH with hyponatremia. Paroxetine: anticholinergic effects (dry mouth, constipation), somnolence are common.	SSRIs are all substrates of various CYP enzymes and subject to drug interactions. Concurrent use with MAOIs, linezolid and methylene blue is contraindicated due to increased risk of serotonin syndrome. Use caution if combining with other serotonergic agents. Increased risk of GI bleeding with NSAIDs, antiplatelet agents. Avoid concurrent use with drugs associated with prolonged QTc interval/torsades de pointes. Paroxetine: potent CYP2D6 inhibitor.	May take 2–3 months for maximum effect. Discontinue gradually.

Drug/Cost[a]	Dosage	Adverse Effects	Drug Interactions	Comments
sertraline Zoloft, generics < $30	Initial: 50–100 mg/day PO Usual: 200 mg/day PO Maximum: 200 mg/day PO; higher doses may be required (up to 400 mg/day)[37][b]	Upon initiation: GI upset, anorexia, dry mouth, diaphoresis, headache, dizziness, insomnia, somnolence, anxiety, agitation, tremor. Usually resolve after 2 wk of therapy. Others: sexual dysfunction, weight gain, SIADH with hyponatremia.	SSRIs are all substrates of various CYP enzymes and subject to drug interactions. Concurrent use with MAOIs, linezolid and methylene blue is contraindicated due to increased risk of serotonin syndrome. Use caution if combining with other serotonergic agents. Increased risk of GI bleeding with NSAIDs, antiplatelet agents. Avoid concurrent use with drugs associated with prolonged QTc interval/torsades de pointes. Sertraline: weak CYP2D6 inhibitor.	May take 2–3 months for maximum effect. Discontinue gradually.

[a] Cost of 30-day supply of mean dosage; includes drug cost only.
[b] If doses higher than recommended are required, carefully screen patients for risk of developing torsades de pointes, e.g., congenital long QT syndrome, CHF, bradyarrhythmia, hypokalemia or hypomagnesemia.

Abbreviations:

CHF = congestive heart failure; GI = gastrointestinal; MAOI = monoamine oxidase inhibitor; NSAID = nonsteroidal anti-inflammatory drug; SIADH = syndrome of inappropriate antidiuretic hormone; SSRI = selective serotonin reuptake inhibitor

Table 4: Drugs for Second- and Third-Line Management of Obsessive-Compulsive Disorder[7]

Drug/Cost[a]	Dosage	Adverse Effects	Drug Interactions	Comments
Drug Class: Alpha-2 receptor antagonists				
mirtazapine Remeron, Remeron RD, generics < $25	Initial: 30 mg QHS PO Usual: 30–60 mg QHS PO[42]	Somnolence, increased appetite/weight gain, dizziness.	Do not use with MAOIs. Additive sedation with other CNS depressants such as alcohol, benzodiazepines; substrate of CYP1A2, 2D6 and 3A4—caution with inhibitors or inducers of these isoenzymes.	**Second-line** monotherapy if SSRIs and augmentation strategies fail. Orally disintegrating tablets can be taken without water.
Drug Class: Serotonin-norepinephrine reuptake inhibitors				
venlafaxine Effexor XR, generics < $25	Initial: 37.5–75 mg/day PO Usual: 112.5–225 mg/day PO Maximum: 300 mg/day PO[41]	Nausea, sleep disturbance, drowsiness, nervousness, dizziness, dry mouth. Dose-related hypertension occurs rarely, particularly at doses ≥225 mg/day.	Use with MAOIs may lead to potentially fatal reaction initially presenting with tremor, agitation, hypomania, hyperthermia and/or hypertension. Inhibitors of CYP2D6 or CYP3A4 may increase venlafaxine levels.	**Second-line** monotherapy if SSRIs fail.
Drug Class: Norepinephrine and serotonin reuptake inhibitors				
clomipramine Anafranil, generics $25–50	Initial: 25 mg/day PO; increase dose Q3–5 days as needed/tolerated Usual: 100–200 mg/day PO Maximum: 250 mg/day PO	CNS effects (agitation on initiation of therapy, confusion, drowsiness, headache), anticholinergic effects (dry mouth, blurred vision, constipation), weight gain, nausea, cardiovascular effects (tachycardia, arrhythmias, orthostatic hypotension), anorgasmia,	May increase effect of anticholinergic drugs, CNS depressants, warfarin; do not use MAOIs concurrently.	**Second-line** monotherapy if SSRIs fail.

Drug/Cost[a]	Dosage	Adverse Effects	Drug Interactions	Comments
		erectile dysfunction.		
Drug Class: Glutamate-modulating drugs				
topiramate Topamax, generics $25–50	Initial: 25 mg daily PO Increase by 25 mg/day every 1–2 wk Range: 50–200 mg/day[52]	CNS effects (e.g., dizziness, ataxia, tremor, sedation, cognitive impairment), GI symptoms (e.g., nausea, dyspepsia, constipation), weight loss (can be beneficial in some patients), metabolic acidosis, nephrolithiasis. Possible increased risk of oral clefts if used during the first trimester of pregnancy.	Additive depressant effects with other CNS depressants. May decrease effectiveness of oral contraceptives; use oral contraceptives containing at least 35 mcg estrogen and add barrier contraceptive protection (condoms). Inhibitors of CYP2C19 may increase topiramate levels, e.g., SSRIs, isoniazid, omeprazole, moclobemide. Phenytoin and carbamazepine can decrease topiramate levels.	**Third-line** augmentation of SRI. May cause acute myopia, with consequent angle closure glaucoma that responds to drug discontinuation; advise patients to consult an ophthalmologist or emergency room *immediately* if they have acute painful/red eyes or decreased/blurred vision. Warn patients about CNS depressant effects; possible risk associated with driving, other hazardous activities.
lamotrigine Lamictal, generics < $25	Initial: 25 mg QHS × 2 wk then 50 mg QHS × 2 wk then 100 mg QHS × 1 wk then 200 mg QHS Range: 100–200 mg/day PO[79]	Benign rash, dizziness, drowsiness, Steven-Johnson syndrome (rare).	Addition of hormonal contraceptives may reduce lamotrigine serum levels by up to 50%. Lamotrigine may increase risk of toxicity of carbamazepine. Valproic acid may increase lamotrigine serum concentration. Phenytoin and carbamazepine may decrease serum concentration of lamotrigine.	**Second-line** augmentation of SRI.
Drug Class: Dopamine receptor partial agonists				
aripiprazole Abilify, generics $50–75	Initial and usual dose: 10–15 mg daily PO[80]	EPS (akathisia, parkinsonism), dizziness, headache, GI complaints, nasopharyngitis, tremor, sedation, insomnia, somnolence, weight gain, lipid and glucose dysregulation.	Carbamazepine (or other strong inducers of CYP2D6 or CYP3A4 such as phenytoin, rifampin) can decrease aripiprazole levels significantly. Ketoconazole, quinidine, fluoxetine or paroxetine (or other strong inhibitors of CYP2D6 or CYP3A4) can increase levels substantially.	**First-line** augmentation of SRI. Advise patients about antipsychotic-associated body temperature dysregulation and prevention of heat stroke, e.g., hydration, sun protection.
Drug Class: Dopamine/serotonin receptor antagonists				

Drug/Cost[a]	Dosage	Adverse Effects	Drug Interactions	Comments
risperidone Risperdal, generics < $25	Initial: 0.5 mg daily PO Usual: Increase dose by 0.5–1 mg Q3–7 days as needed/tolerated to a maximum of 3 mg/day[80]	Somnolence, headache, weight gain, lipid and glucose dysregulation, orthostatic hypotension, rhinitis, anxiety, dose-related hyperprolactinemia, EPS and QT_c prolongation. Risk of intraoperative floppy iris syndrome in patients undergoing cataract surgery who have been exposed to risperidone.	Additive sedation with CNS depressants; may potentiate antihypertensive drug effects; inhibitors of CYP3A4, such as clarithromycin, erythromycin, grapefruit juice, ketoconazole or prednisone, may increase risperidone levels; inducers of CYP3A4 such as carbamazepine, phenytoin or rifampin may decrease risperidone levels.	**First-line** augmentation of SRI. Advise patients about antipsychotic-associated body temperature dysregulation and prevention of heat stroke, e.g., hydration, sun protection.
Drug Class: N-methyl-D-aspartate Receptor Antagonists				
memantine Ebixa, generics $75–100	Initial: 10 mg daily PO x 1 wk Usual: 20 mg daily in 1–2 doses[81]	Generally well tolerated; dizziness, headache, confusion, constipation, nausea/vomiting.	Not affected by cytochrome P450 system. Theoretically, urinary alkalizers such as carbonic anhydrase inhibitors may decrease the clearance of memantine.	**Second-line** augmentation of SRI.
amantadine generics < $25	100 mg daily PO[49]	Nausea, constipation, dry mouth, insomnia, anxiety, impaired concentration, livedo reticularis, orthostatic hypotension, ankle edema.	Not affected by cytochrome P450 system. Theoretically, urinary alkalizers such as carbonic anhydrase inhibitors may decrease the clearance of memantine.	**Third-line** augmentation of SRI. Risk of parkinsonism-hyperpyrexia syndrome with abrupt discontinuation; taper gradually.

[a] Cost of 30-day supply of mean dosage, unless otherwise specified; includes drug cost only.

Dosage adjustment may be required in renal impairment; see Dosage Adjustment in Renal Impairment.

Abbreviations: CNS = central nervous system; EPS = extrapyramidal symptoms; GI = gastrointestinal; MAOI = monoamine oxidase inhibitor; SRI = serotonin reuptake inhibitor (includes SSRIs, venlafaxine and clomipramine); SSRI = selective serotonin reuptake inhibitor

Suggested Readings

Katzman MA, Bleau P, Blier P et al. Canadian clinical practice guidelines for the management of anxiety, posttraumatic stress and obsessive-compulsive disorders. *BMC Psychiatry* 2014;14(Suppl 1):S1.

Pittenger C, Brennan BP, Koran L et al. Specialty knowledge and competency standards for pharmacotherapy for adult obsessive-compulsive disorder. *Psychiatry Res* 2021;300:113853.

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